PAPER ID-310533

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BTECH
(SEM V) THEORY EXAMINATION 2025-26
POWER SYSTEM-I

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data, choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Explain Average load, Demand Factor in a power plant.	1	K2
b.	Compare conventional and non-conventional energy sources with suitable examples.	1	K2
c.	Why the line capacitance is neglected in short transmission lines.	2	K2
d.	Explain Ferranti effect and how it can be reduced?	2	K2
e.	A span of 200 m carries a conductor (weight = 0.9 kg/m). Find sag if tension = 2000 kg.	3	K2
f.	Describe the uses and advantages of bundled conductors.	4	K2
g.	Write the differences between underground cables and overhead transmission line.	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	A generating station has a maximum demand of 35 MW, a load factor of 50 %, a plant capacity factor of 80 % and a plant use factor of 62 %. Find (i) the reserve capacity of the plant (ii) the daily energy produced and (iii) Maximum energy that could be produced daily if the plant while running as per schedule, were fully loaded.	1	K2
b.	Derive the A, B, C, D constants for the transmission line represented by nominal T section and draw its phasor diagram.	2	K2
c.	Explain the calculation procedure of maximum sag of a transmission line at equal supports including wind pressure loading.	3	K2
d.	Show that the inductance per unit length of an overhead line due to internal flux linkages is constant and is independent of size of conductor.	4	K2
e.	Discuss the inter-sheath grading of cables. What are practical difficulties in the grading of cables.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain in detail the nuclear reaction theory and working principle of a nuclear power station with plant layout.	1	K2
b.	Define and explain the importance of the following terms in generation: (i) connected Load (ii) Maximum Demand (iii) Diversity Factor (iv) Average Load.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	An overhead transmission line operates at 220 kV between phases at 50 Hz. The conductors are arranged in a 5 m delta formation. Calculate the maximum diameter of the conductor that can be used for no corona loss under fair weather conditions. Assume an air density factor of 0.95 and an irregularity factor of 0.85. The critical voltage is 230 kV. Also, find the power loss under stormy weather conditions.	2	K2
b.	What is the Corona Effect**?** Discuss factors affecting corona and its advantages/disadvantages.	2	K2

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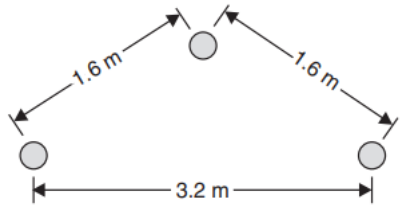
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5. Attempt any *one* part of the following: 07 x 1 = 07

a.	Each conductor of a 33 kV, 3-phase system is suspended by a string of three similar insulators, the capacitance of each disc is nine times the capacitance to ground. Calculate the voltage across each insulator. Determine the string efficiency also.	3	K2
b.	A transmission line has a span of 150 m between level supports. The line conductor has a cross-sectional area of 1.25 cm^2 and it Weighs 120kg per 100m. If the breaking stress of the copper conductor is 4220 kg/ cm^2. Calculate the maximum sag for a safety factor of 4. Assume maximum wind pressure of 90 kg/m^2 of projected surface.	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Determine the inductance of a 3-phase line operating at 50 Hz and conductors arranged as follows. The conductor diameter is 0.8 cm. 	4	K2
b.	Two conductors of a single-phase line, each of 1 cm diameter, are arranged in a vertical plane with one conductor mounted 1 m above the other. A second identical line is mounted at the same height as the first and spaced horizontally 0·25 m apart from it. The two upper and the two lower conductors are connected in parallel. Determine the inductance per km of the resulting double circuit line.	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	A single core cable has an inner diameter of 5 cms and a core diameter of 1.5 cm. Its paper dielectric has a working maximum dielectric stress of 60 kV/cm. Calculate the maximum permissible line voltage when such cables are used on a 3-phase power system.	5	K2
b.	Explain the grading method of underground cables with mathematical expressions.	5	K2

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